



Certificate & Results Management System

Tamper-proof certificates · semester-wise results · public verification · role-based admin portal



Role-Based Access



Blockchain Anchoring

Bulk Upload



Audit Trail



Dark / Light Themes



Docker + PM2



Contents

1 — Overview & Key Features

2 — Roles & Permissions

3 — Technology Stack

4 — Public Website (4 screens)

5 — Admin Portal (12 screens)

6 — Access Details



Overview

A full-stack university platform that lets the records office issue cryptographically tamper-proof certificates, manage semester-wise student results, and let the public verify credentials — all behind a secure, role-based admin portal with a complete audit trail.



Public

Verify a certificate by hash · check a student result · raise an issue — no login required, no PII leaked.

**Admin Portal**

Dashboard, certificates, results, analytics, issues, audit log, user management, branding and downloads — with 7 roles.

Key Features

- SHA-256 Tamper-Proof Certificates**
Every certificate gets a unique cryptographic hash; optional anchoring to the Sepolia testnet blockchain and IPFS.
- Role & Department Scoping**
Seven roles with fine-grained section permissions; normal users can be restricted to specific departments and cannot bypass the filter.
- Bulk Upload With Mismatch Reports**
CSV / XLSX uploads for certificates and results with header detection and a full mismatch report before committing.
- Public Verification, Privacy First**
Anyone can verify a hash; the API never exposes student personal data to the public.
- Complete Audit Trail**
Every administrative action is logged with search; session tokens survive restarts and auto-rotate on idle.
- Python-Fast Production Readiness**
Express + PostgreSQL JSONB backend, React SPA frontend, PM2 process management and Docker Compose deployment.

Roles & Permissions

ROLE	DEFAULT ACCESS
Admin	Full access to every section and all departments.
Normal User	Section-based access; page permissions controlled per user.

ROLE	DEFAULT ACCESS
Department Head	Certificates, Results, Verify, Analytics, Issues, Activity (department-scoped).
Privacy Manager	Verify, Issues, Activity, Downloads (audit & privacy oversight).
Cyber Security	Campus Monitor, Verify, Issues, Activity.
HR	Certificates, Results, Verify, Downloads, Activity.
Legal	Verify, Issues, Activity, Downloads.

Normal users can be limited to specific departments at the API level (403 enforced) — empty department selection means all departments.

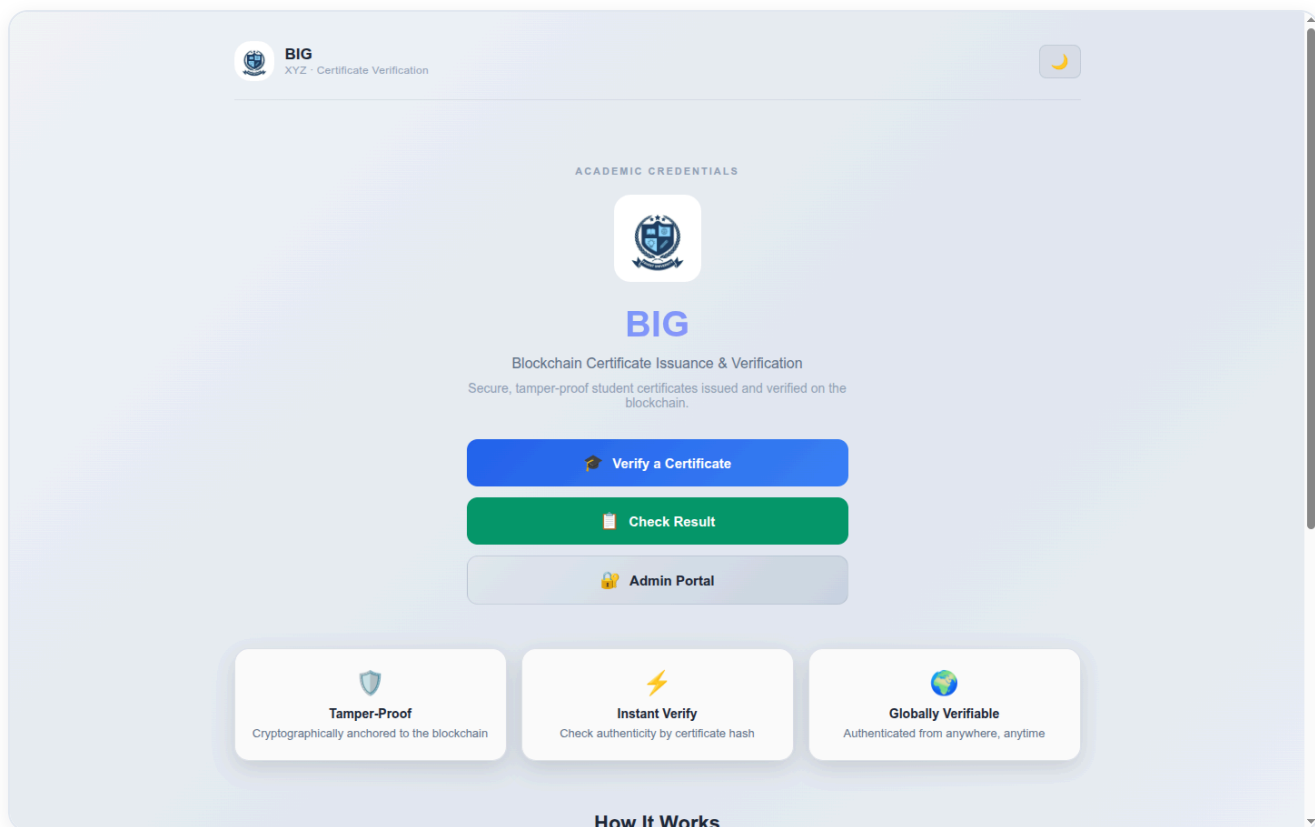
 Technology Stack

FRONTEND	React (Create React App), custom CSS with light/dark themes
BACKEND	Node.js + Express REST API
DATABASE	PostgreSQL 16 (JSONB for subjects, permissions & departments)
BLOCKCHAIN	Optional Sepolia testnet anchoring of certificate hashes
FILES	XLSX parsing, CSV export
OPS	PM2 process manager, Docker Compose

 Application Screens

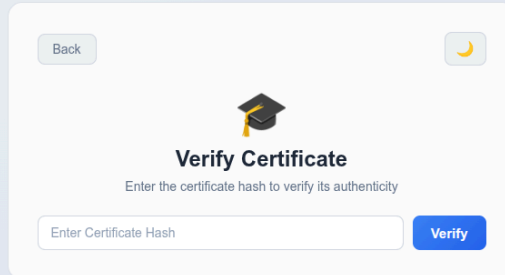
Public Landing Page

Branded landing page with university logo, name, tagline, How It Works explainer (Issue → Store → Verify), trust band and dark/light theme toggle.



Verify Certificate (Public)

Anyone can verify a certificate by entering its hash. The backend returns only validity — no student personal data is exposed.



The image shows a mobile application interface for verifying a certificate. It features a white card with rounded corners on a light blue background. At the top left of the card is a 'Back' button, and at the top right is a moon icon. In the center, there is a graduation cap icon above the text 'Verify Certificate'. Below this, a subtitle reads 'Enter the certificate hash to verify its authenticity'. At the bottom, there is a text input field with the placeholder 'Enter Certificate Hash' and a blue 'Verify' button to its right.

Check Result (Public)

Students can look up semester-wise results by roll number, and raise issues against them without any login.

[Back](#)



Verify Result

Secure login with Aadhaar number — an OTP will be sent to your registered mobile

[Send OTP](#)

Admin Portal — Login

Secure login with script-hashed passwords, rate-limited attempts, and 64-char session tokens stored server-side.

[Back](#)



Admin Login

Enter admin credentials to access the portal

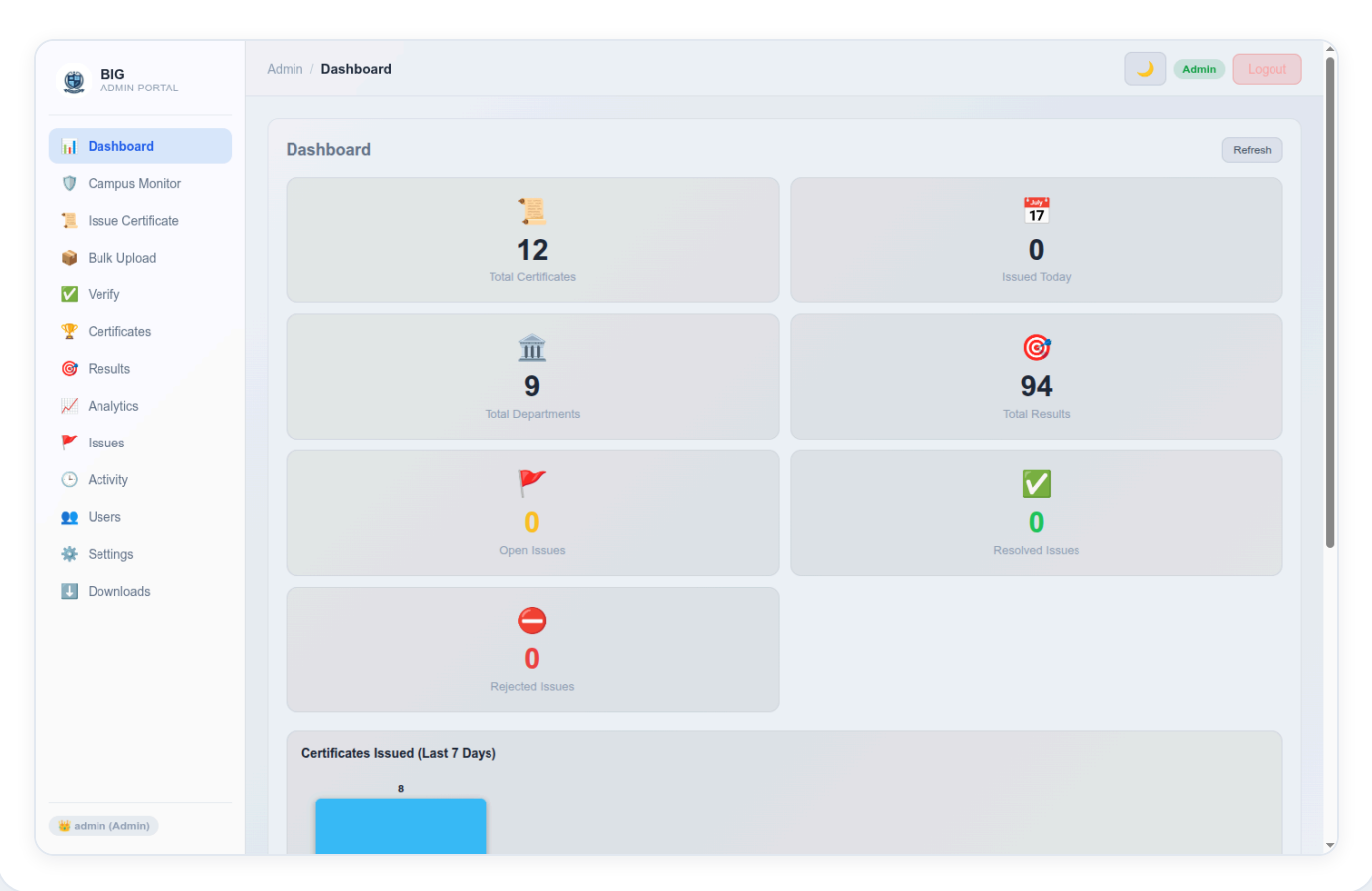
Username

Password

Login

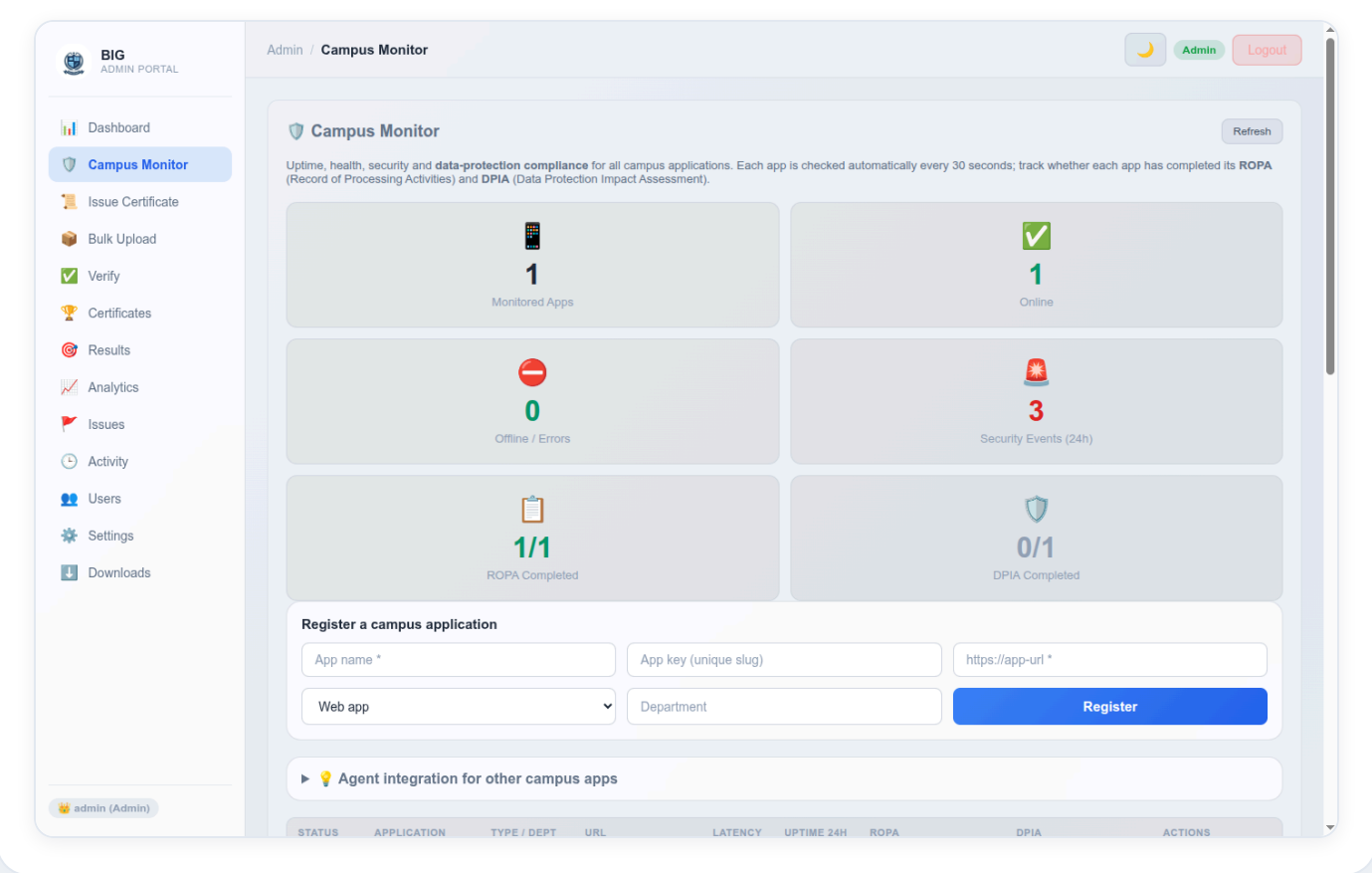
Dashboard

Live statistics: total certificates, issued today, departments, results and issue counts, plus 7-day / department-wise / semester-wise charts.



Campus Monitor

Public-facing monitors ships with live certificate issuance feed and uptime/health status.



Issue Certificate

Issue a single certificate with name, roll, course, department, year and email; generated SHA-256 hash with optional IPFS and on-chain (Sepolia) anchoring.

Dashboard

Campus Monitor

Issue Certificate

Bulk Upload

Verify

Certificates

Results

Analytics

Issues

Activity

Users

Settings

Downloads

admin (Admin)

Admin / Issue Certificate

Issue Certificate

Student Name *

e.g. John Doe

Roll Number *

e.g. 2021001

Course

e.g. B.Tech

Department

e.g. Computer Science

Year

e.g. 2024

Email

e.g. student@xyz.edu

Certificate File (PDF/Image)

Choose file

No file chosen

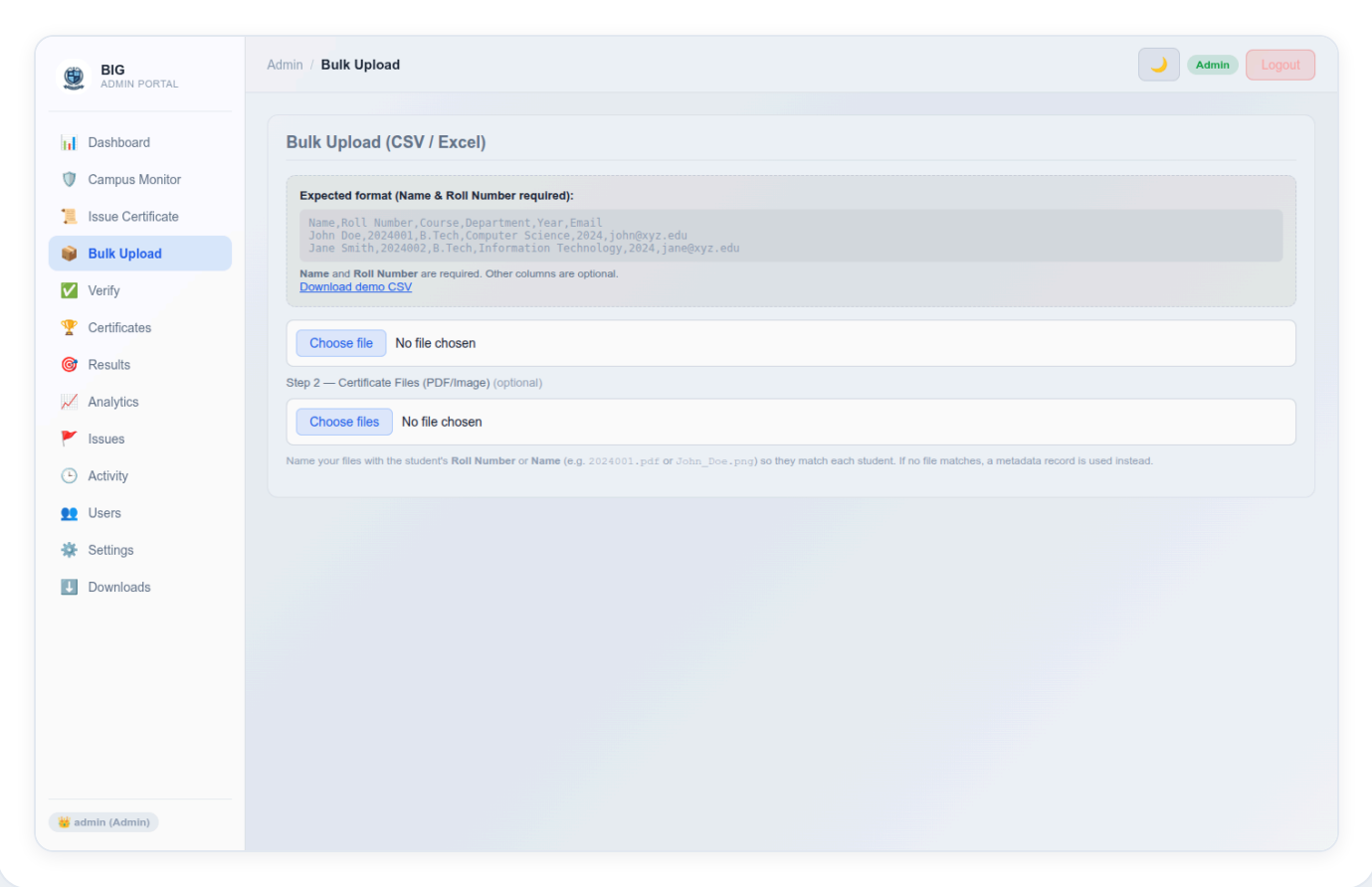
Add Certificate to Blockchain

Bulk Upload

Upload certificates or results as CSV/XLSX with header detection and a mismatch report; processing is gated until the data is complete.

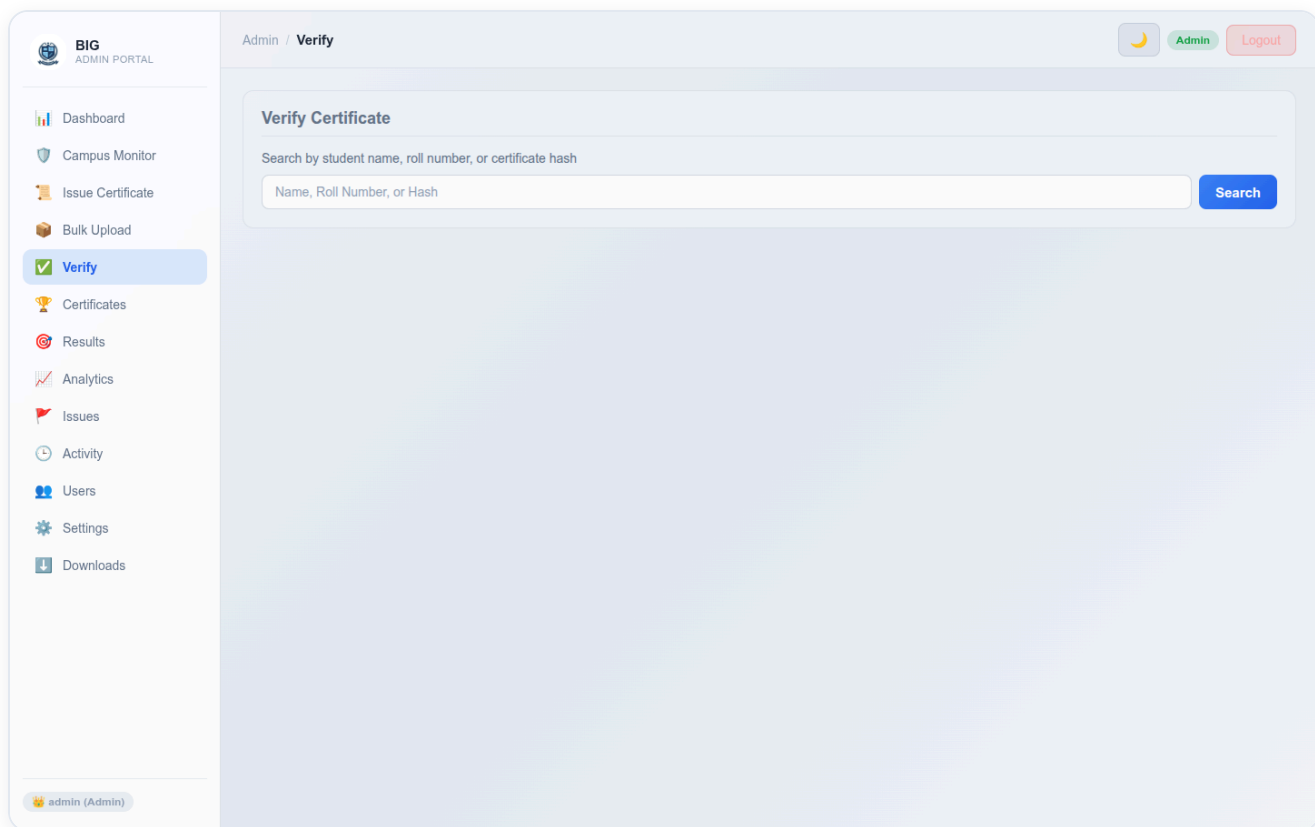
file:///home/pradeep/certificate-app/presentation/certificate-app-presentation.html

10/20



Verify (Admin)

Admin-facing verification for certificates and results, with immediate pass/fail feedback.



Certificates Registry

Searchable registry by hash / name / roll / email, on-chain status badges, and CSV / XLSX export.

Dashboard

Campus Monitor

Issue Certificate

Bulk Upload

Verify

Certificates

Results

Analytics

Issues

Activity

Users

Settings

Downloads

Admin / Certificates

Admin

Logout

All Certificates

Refresh

Search by name, roll no, or email

All Courses

#	NAME	ROLL NO	COURSE	DEPARTMENT	YEAR	IPFS	ACTION
1	Anupam	-	-	-	-	View	Delete
2	ram	-	-	-	-	View	Delete
3	tushar	-	-	-	-	View	Delete
4	ketan	-	-	-	-	View	Delete
5	ChatGPT Image May 28, 2026, 10_39_23 AM	-	-	-	-	View	Delete
6	Incident Responses	-	-	-	-	View	Delete
7	microsoft	-	-	-	-	View	Delete
8	Board Resolution_Bankacopening	-	-	-	-	View	Delete
9	Parham Risk Management - BasicUserGuide 0526	-	-	-	-	View	Delete
10	.-lock.Parham Risk Management - BasicUserGuide 0526	-	-	-	-	View	Delete
11	vivek	-	-	-	-	View	Delete
12	Anupam xyz	-	-	-	-	View	Delete

Showing 12 of 12 certificates

admin (Admin)

Results

Semester-wise student results with multiple subjects per record; unique roll + semester constraint prevents duplicates.

Dashboard

Campus Monitor

Issue Certificate

Bulk Upload

Verify

Certificates

Results

Analytics

Issues

Activity

Users

Settings

Downloads

admin (Admin)

Admin / Results

Semester Results

Store and view student marks per semester, organized by department.

Bulk Upload Results

Upload a CSV / Excel with one row per subject. Columns: Roll Number , Name , Department , Semester , Subject , Marks , Max Marks , Grade . Rows with the same student + semester are grouped into one result.

Expected format (Roll Number & Subject required):

Roll Number ,Name,Department,Semester,Subject,Marks,Max Marks,Grade
2024001,John Doe,Computer Science,Sem 1,Mathematics,85,100,A
2024001,John Doe,Computer Science,Sem 1,Physics,78,100,B
2024002,Jane Smith,Information Technology,Sem 2,Data Structures,72,100,B

[Download demo CSV](#)

Choose file No file chosen

Save Bulk (0 rows)

Add Semester Result

Student Name *

e.g. John Doe

Roll Number *

e.g. 2024001

Aadhaar Number (for result login)

12-digit Aadhaar (optional)

Registered Mobile (for OTP)

10-digit mobile (optional)

Department *

Select department...

Semester *

Select semester...

Subjects & Marks

+ Add Subject

SUBJECT	MARKS	MAX MARKS	GRADE
Subject	Marks	Max	Grade

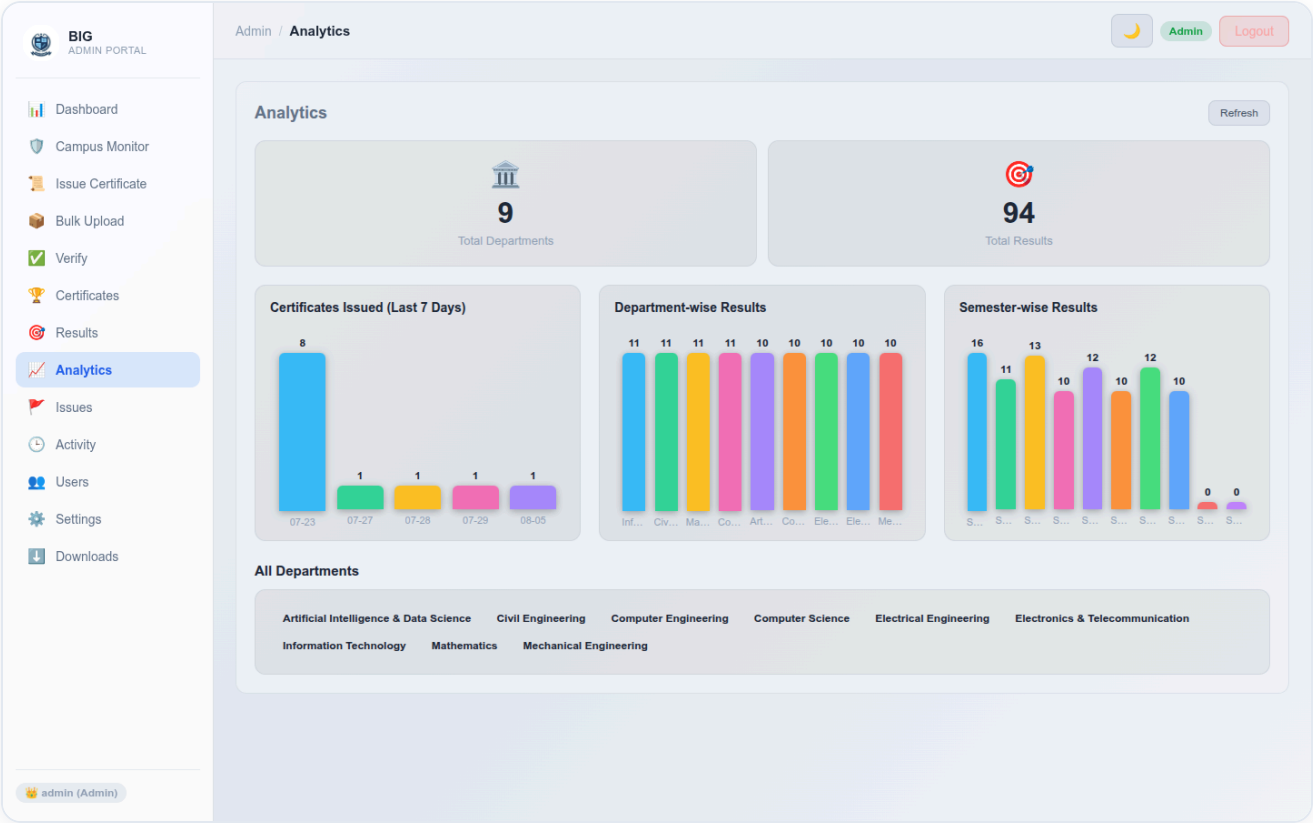
Save Result

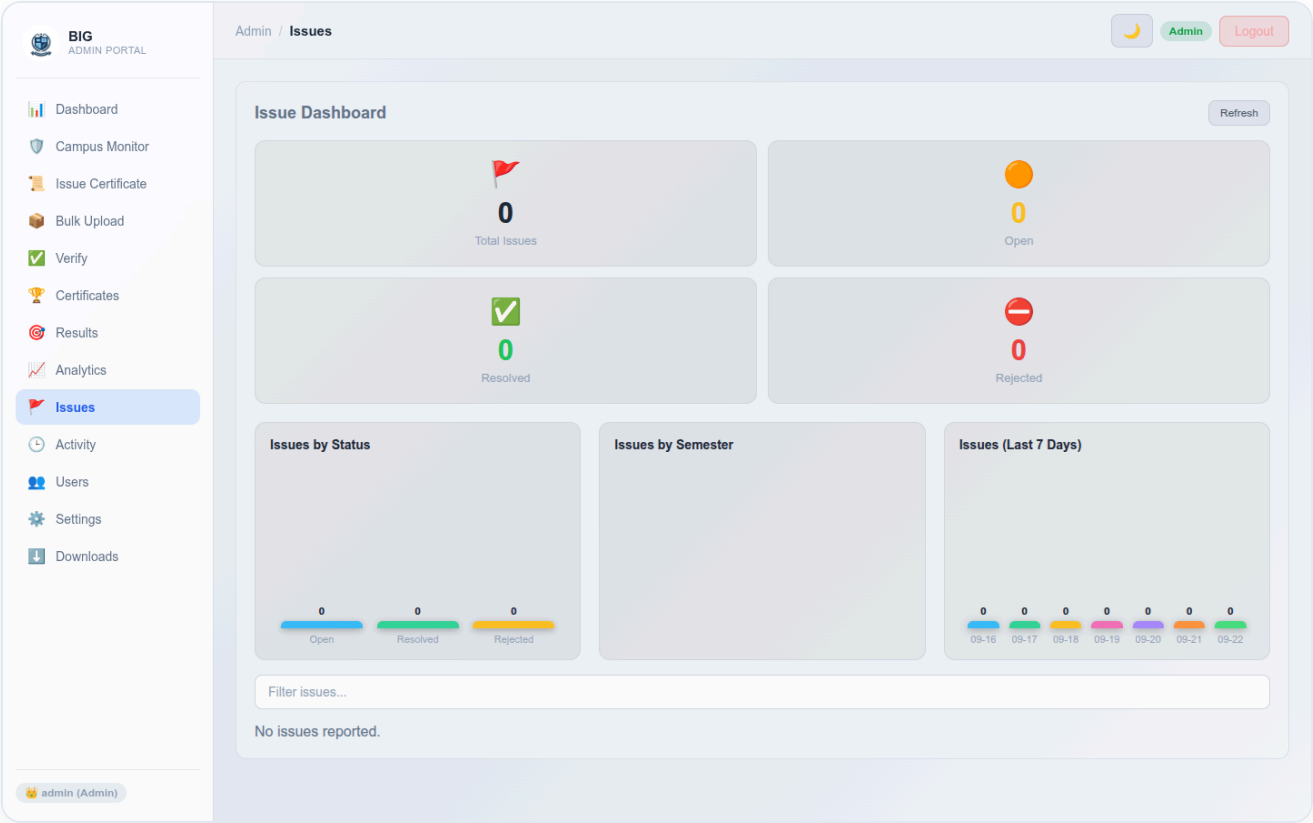
Analytics

Insights into certificates and results across semesters, departments and board state.

file:///home/pradeep/certificate-app/presentation/certificate-app-presentation.html

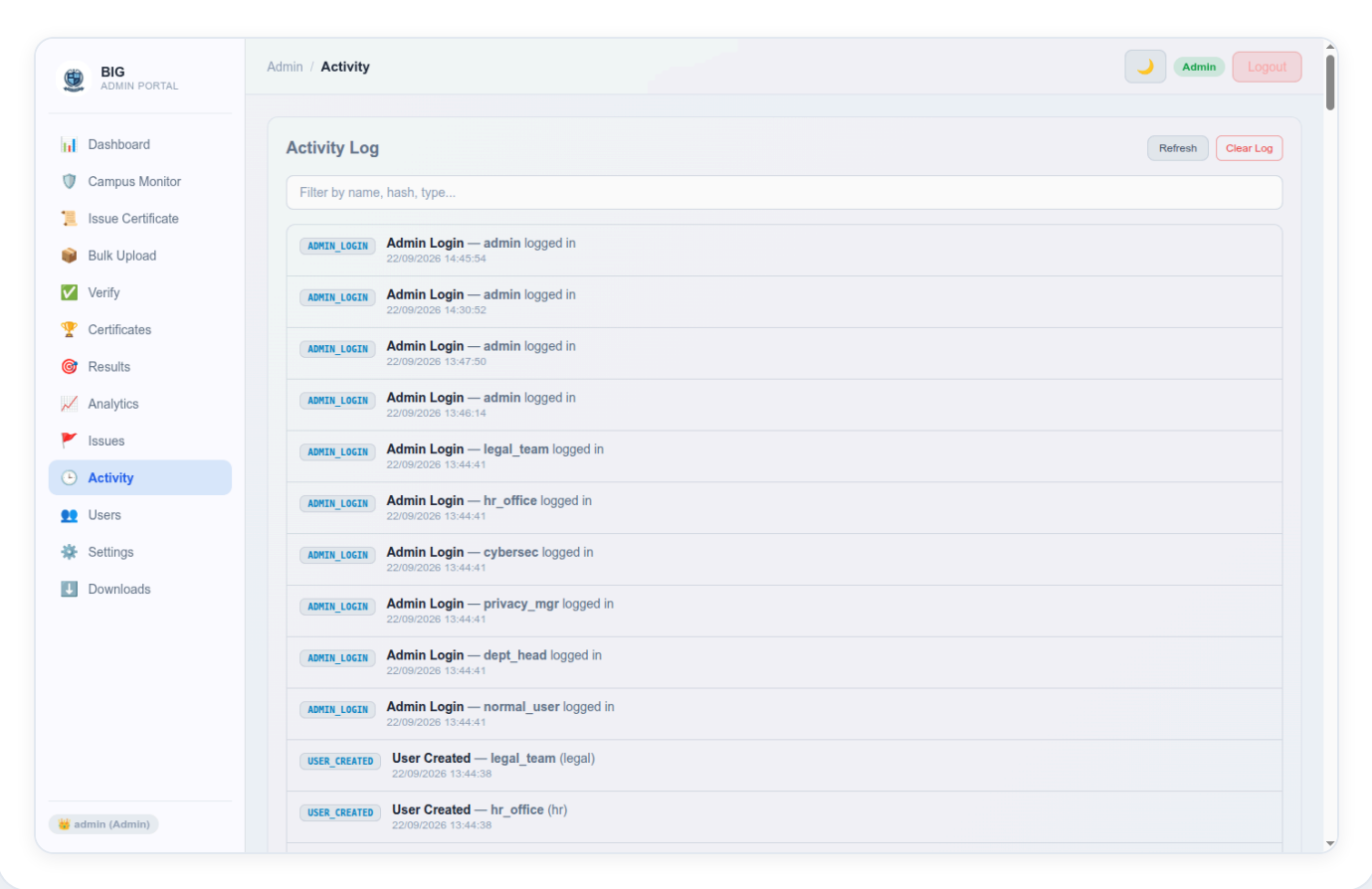
14/20





Activity Log

Full audit trail of every key action (logins, issuances, deletions, user and brand changes) with search.



User Management

Role-based accounts: Admin, Normal User, Department Head, Privacy Manager, Cyber Security, HR and Legal — with per-user section permissions and department scoping.

Dashboard

Campus Monitor

Issue Certificate

Bulk Upload

Verify

Certificates

Results

Analytics

Issues

Activity

Users

Settings

Downloads

admin (Admin)

Admin / Users

Admin

Logout

Refresh

User Management

Create login accounts and manage passwords for university staff.

Create New User

Username

Password (min 4 chars)

Normal User

Create User

Section Access:

Select All

Clear

Default

☒ Issue Certificate

☒ Bulk Upload

☒ Verify

☒ Certificates

☐ Results

☐ Analytics

☐ Campus Monitor

☐ Issues

☒ Activity

☒ Downloads

Department Access:

Select All

Clear

Leave empty for all departments

☐ Artificial Intelligence & Data Science

☐ Civil Engineering

☐ Computer Engineering

☐ Computer Science

☐ Electrical Engineering

☐ Electronics & Telecommunication

☐ Information Technology

☐ Mathematics

☐ Mechanical Engineering

Search by username, role, access, or department...

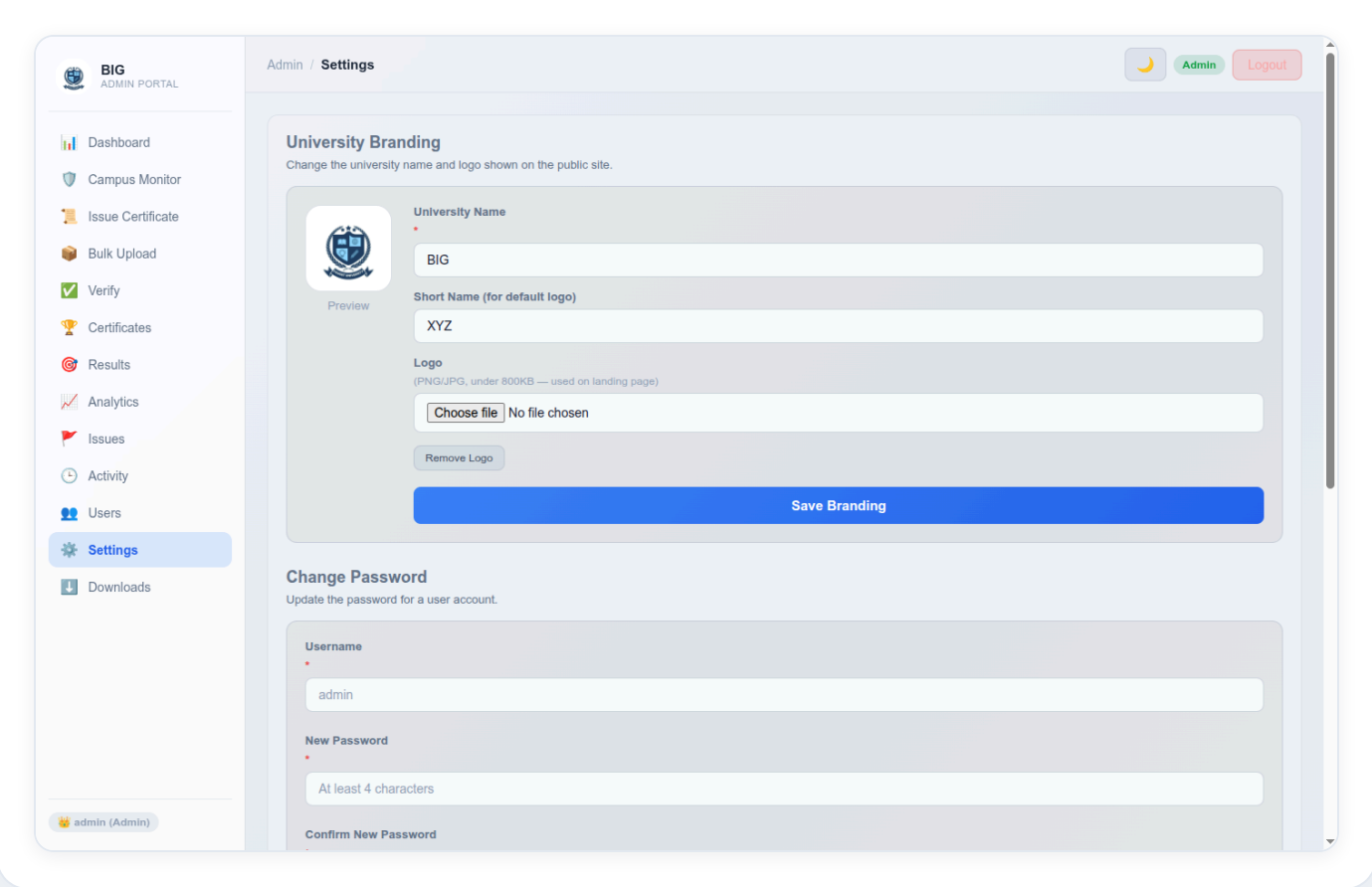
USERNAME	ROLE	ACCESS	CREATED	CHANGE PASSWORD	ACTION
Anupam	CYBER SECURITY Cyber Security	Campus Monitor Verify Issues Activity Dept: 1	24/08/2026	Set Password	Delete
normal_user	NORMAL USER Normal User	Issue Certificate Bulk Upload Verify Certificates Activity Downloads Dept: All	22/09/2026	Set Password	Delete
dept_head	DEPARTMENT HEAD Department Head	Certificates Results Verify Analytics Issues Activity Dept: All	22/09/2026	Set Password	Delete
admin (you)	ADMIN	All sections	07/08/2026	Set Password	Delete
privacy_mgr	PRIVACY MANAGER Privacy Manager	Verify Issues Activity Downloads Dept: All	22/09/2026	Set Password	Delete

Settings

University branding (name, short name, logo) and self-service password change.

file:///home/pradeep/certificate-app/presentation/certificate-app-presentation.html

18/20



Downloads

One-stop export hub for the certificate registry and results as CSV or Excel.

Dashboard

Campus Monitor

Issue Certificate

Bulk Upload

Verify

Certificates

Results

Analytics

Issues

Activity

Users

Settings

Downloads

Admin / Downloads

Download Certificate List

Download CSV

Download Excel

admin (Admin)

Access Details

USER	ROLE	PASSWORD
admin	Admin	admin123
normal_user	Normal User	staff123
dept_head	Department Head	staff123
privacy_mgr	Privacy Manager	staff123
cybersec	Cyber Security	staff123
hr_office	HR	staff123
legal_team	Legal	staff123